MIT-WPU | Faculty of Engineering and Technology

School of Computer Science and Engineering
Department of Computer Engineering and Technology

S. Y. B. Tech CSE **AY 2026-27** **Semester: 3**

Computer Networks Lab (CSE40110)

Assignment No.: 7

Title: DHCP, DNS, Web Server and Email configuration in cisco packet tracer.

Aim: DHCP, DNS and Web Server configuration (Use: Packet Tracer/GNS). Configure network using Dynamic Host Configuration Protocol (DHCP), DNA and Web Server Use Ping utility to test connectivity.

Objectives: To learn the DHCP, DNS and Web Server installation and understand their practical usage.

Theory: (Handwritten & Scanned)

- Student's should write One-page theory report (Handwritten & Scanned)
  - Dynamic Host Configuration Protocol (DHCP) (Functionality and need)
  - DNS (Domain Name System resource records, iterative and recursive lookups)
  - SMTP, POP3 and IMAP (Functionality and port numbers)

- **Design Screenshots:**

  Paste design screen shots

- **Output Screenshots:**

  Student's should write/paste Code here (Screenshots/Snaps are not accepted)

- **Students Observation:**

  Write your observations on the mentioned objective

FAQ (Handwritten & Scanned)

1. What are the different DHCP messages involved in the dynamic IP address allocation process?
2. Explain the DORA process used by DHCP for dynamic IP address allocation.
3. What are the different types of Resource Records (RRs) used in DNS? Explain their purpose.
4. Compare SMTP, POP3, and IMAP based on their functions and methods of handling email messages.